INCH-POUND
MIL-DTL-23053/15B
17 June 1996
SUPERSEDING
MIL-I-23053/15A
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE POLYOLEFIN, HEAVY-WALL, COATED, FLEXIBLE, OUTER WALL CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

Polymer type: The base polymer used in formulating the outer wall of this sleeving shall be a polyolefin.

Continuous operating temperature range: -55°C (-67°F) to +110°C (+230°F).

Color: The heat shrinkable sleeving shall be furnished in colors that conform to the requirements of Class II of MIL-STD-104. Black shall be the standard acquisition color.

Longitudinal change: +1 -10 percent

Classes: The sleeving shall be furnished in the following classes, as specified:

Class 1 - Heavy Wall
Class 2 - Medium Wall

Coatings: The inner wall of the sleeving shall be coated with an adhesive or sealant type material which shall effectively seal the enclosed area against moisture.

Adhesive: Adhesive is for pressure retention and load bearing applications.

Sealant: Sealant is for water hold out applications.

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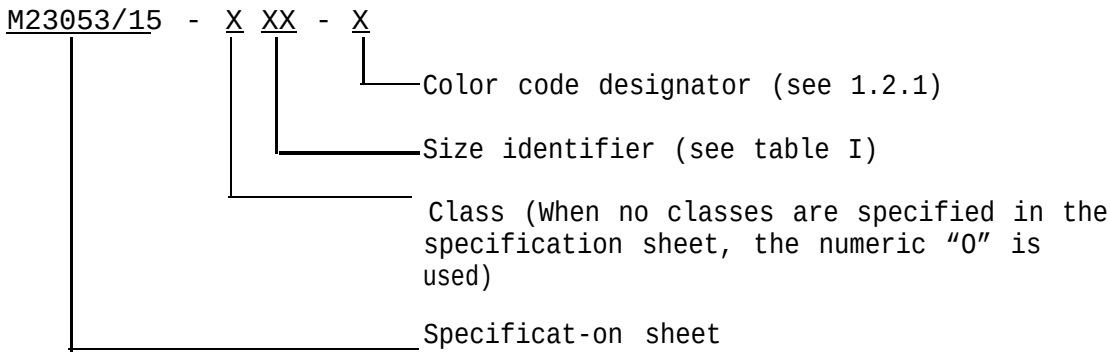
Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers shown as follows:



Example: Class 1, black sleeving, 3.00 inch (76.2 mm) as supplied ID shall be identified as M23053/15-105-0.

TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum 3/	After unrestricted shrinkage	
		ID maximum	Wall thickness nominal
Class 1			
M23053/15-101-0	.750(19.1)	.220(5.59)	.105(2.7)
M23053/15-102-0	1.100(27.9)	.375(9.53)	.120(3.1)
M23053/15-103-0	1.500(38.1)	.500(12.7)	.140(3.6)
M23053/15-104-0	2.000(50.8)	.750(19.1)	.155(3.44)
M23053/15-105-0	3.000(76.2)	1.250(31.8)	.155(3.44)
M23053/15-106-0	4.000(101.6)	1.750(44.5)	.155(3.44)
M23053/15-107-0	.300(7.6)	.100(2.5)	.050(1.3)
M23053/15-108-0	.400(10.2)	.150(3.8)	.060(1.5)
M123053/15-109-O	1.300(33.0)	.375(9.5)	.120(3.1)
M123053/15-110-O	1.700(43.2)	.500(12.7)	.155(3.4)
M223053/15-111-0	2.700(68.6)	.900(22.9)	.155(3.4)
M23053/15-112-0	4.500(114.3)	1.750(44.5)	.170(4.3)
Class 2			
M23053/15-201-0	.400(10.2)	.150(3.81)	.060(1.52)
M23053/15-202-0	.750(19.1)	.220(5.59)	.060(1.52)
M23053/15-203-0	1.100(27.9)	.375(9.53)	.105(2.67)
M23053/15-204-0	1.300(33.0)	.375(9.53)	.105(2.67)

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TABLE I. Construction details, inches (mm)). 1/ - Continued

Military part number	As supplied ID minimum 3/	After unrestricted shrinkage	
		ID maximum	Wall thickness nominal
M23053/15-205-0	1.500(38.1)	.500(12.7)	.120(3.05)
M23053/15-206-0	1.700(43.2)	.500(12.7)	.140(3.56)
M23053/15-207-0	2.700(68.6)	.900(22.9)	.155(3.44)
M23053/15-208-0	2.000(50.8)	.750(19.1)	.140(3.6)

1/ Diameter limits for objects to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ As supplied ID is for uncoated product. An allowance of approximately .050 inch (1.3 mm) may be necessary depending on coating.

Unrestricted shrinkage: Test method 4.6.5.2; 200° ± 2°C (392° ± 4°F) for 10 minutes maximum.

TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Low temperature flexibility	No cracking	4.6.7.1 -55° ± 2°C (-67° ± 4°F)
Heat shock	No cracking, flowing or dripping of outer wall	4.6.8 225° ± 3°C (437° ± 6°F)
Secant modulus, psi (MPa), maximum	25,000(172.4)	4.6.12.1, 2 percent strain, ASTM D882
Color stability	Pass	4.6.15 175° ± 2°C (347° ± 4°F) 24 hours

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	1200(8.3)	4.6.13 ASTM D638, 2 inches/minute
Ultimate elongation, percent, minimum	200	4.6.13 ASTM D638, 2 inches/minute
Dielectric strength, volts/roil (Kv/mm), minimum	200(7.9)	4.6.2 ASTM D2671
Volume resistivity, Ohm-cm, minimum	10^{13}	4.6.2 ASTM D876
Corrosion	No corrosion	4.6.10.1 and 4.6.10.2 121° ± 2°C (250° ± 4°F) for 16 hours
Flammability	1/	4.6.14 Procedure C ASTM D2671
Water absorption, percent, maximum	0.5	4.6.2 ASTM D570 24 hrs at 23°C
Heat resistance, properties after:		4.6.9 175° ± 2°C (347° ± 4°F), 168 hours
Ultimate elongation, percent, minimum	100	
Tensile strength, psi (MPa), minimum	1,000(6.9)	
Fluid resistance, properties after:		4.6.11
Tensile strength, psi (MPa), minimum	750 (5.2)	

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Dielectric strength volts/roil (Kv/mm), minimum	200(7.9)	
Fungus resistance	No growth	4.6.2 ASTM G21

1/ Record whether any flaming or glowing particles or flaming drops fall from the specimen. If more than 25 percent of the indicator flag is burned away or charred (brown scorching and soot shall be ignored) after all of the five applications of flame or if flaming or glowing particles or flaming drops fall from the specimen at any time and ignite the cotton (flameless charring of the cotton shall be ignored), the specimen shall be judged to have conveyed flame and shall be considered as failing. In addition, any specimen which continues to flame or glow from one flame application longer than one minute shall not be acceptable.

Storage life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 2 years. Conformance to 3.5. See 3.5.2 for storage life extension.

Flammability: Only the outer jacket is flame retardant.

Intended use: Heat shrinkable, flexible, heavy wall, coated sleeving is used for repair of heavy duty cables, splice covers.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 5970-1129-15)

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